Shih-Yang Lin

Assistant Professor, Department of Economics, National Dong Hwa University

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EDUCATION

National Taiwan University (NTU), Taipei, Taiwan

Ph.D., Economics, 2025

M.A., Economics, 2016

National Sun Yat-Sen University (NSYSU), Kaohsiung, Taiwan

B.A., Political Economy, 2008

Dissertation

"The Role of Financial Frictions in Explaining Macroeconomic Dynamics."

Dissertation Supervisor: Yi-Chan Tsai

RESEARCH FIELDS AND INTERESTS

Fields: Macroeconomics and Econometrics

Interests: Financial frictions, inflation inequality, nonlinear models with endogenous regressors, and inference and design for policy evaluation.

ACADEMIC EMPLOYMENT

National Dong Hwa University, Hualien, Taiwan

Assistant Professor, Department of Economics, 2025–present

Academia Sinica, Taipei, Taiwan

Research Assistant, 2016–2018

RESEARCH GRANTS

National Science and Technology Council Grant (NSTC 115-2410-H-259-059-), 2026–2027

AWARDS AND HONORS

Best Doctoral Dissertation Award (Sho-Chieh Tsiang Theoretical Economics), Taiwan Economic Association, 2025

Doctoral Dissertation Fellowship of the Humanities and Social Sciences, National Science and Technology Council, 2023

Asia Bank Scholarship, National Taiwan University, 2022

Liu Ta-Chung Scholarship, National Taiwan University, 2020

Harold H.C. Han Scholarship, National Taiwan University, 2019

Chu, Cyrus C. Y. and Chen, Tain-Jy Scholarship, National Taiwan University, 2018

PUBLICATION

Refereed Journal Articles

1. **"U.S. Inflation Inequality between 2010 and 2023."** Coauthored with YiLi Chien and Yi-Chan Tsai. 2026. *Federal Reserve Bank of St. Louis Review*, 108: (2), pp. 1-22, doi:10.20955/r.2026.02

2. **"Financial Frictions and Uncertainty Shocks."** Coauthored with Yi-Chan Tsai and Po-Yuan Wang. 2025. *Macroeconomic Dynamics*, 29: e143, doi:10.1017/S1365100525100497

3. **"Estimating the Willingness to Pay for Voting when Absentee Voting is not Allowed."** Coauthored with C.Y. Cyrus Chu and Wen-Jen Tsay. 2021. *Social Science Quarterly*, 102: (4), pp. 1380-1393, doi:10.1111/ssqu.13014

Non-refereed Journal Articles

1. **"Constructing the DSGE-VAR Model — Taiwan's Medium-to-Long-Term Economic Growth Rate Forecast."** Coauthored with Yi-Chan Tsai, Ruey Yau, and Kuo-Hsuan Chin, 2022. *Quarterly Bulletin, Central Bank of the Republic of China (Taiwan)*, 44: (2), pp. 27-70.

WORKING PAPER

1. **"Consumption Weighting and Inflation Inequality across Household Groups: Evidence from Malawi."** Coauthored with Angela Seyama and Chia-Yu Hung. *Under review*.

   Households with different consumption baskets can face different inflation exposure even under common national category price changes, which matters when transfers and poverty lines are adjusted using a single index. Using three waves of Malawi's Integrated Household Survey, we compare six household-group contrasts over a moderate-inflation period in 2016–2019 and a high-inflation period in 2019–2024 under equal-household democratic and expenditure-weighted plutocratic aggregation. Under the household-oriented democratic measure, bottom-quintile, rural, below-primary-education, and female-headed households have higher exposure in both periods. Plutocratic aggregation widens all four gaps because higher-spending households within groups generally hold baskets exposed to slower price growth, and this gradient is weaker among the more exposed groups. It also reverses the age comparison in both periods. Food accounts for most of the expenditure, residence, and education gaps, while housing, water and energy is central to female headship gap. Thus, the main socioeconomic pattern is robust, but gap magnitudes and some demographic conclusions depend on how households are weighted. The aggregation rule should therefore be stated when reporting group inflation. doi:10.21203/rs.3.rs-10878722/v1

2. **"The Labor Supply of Married Women and Spousal Tax Deductions in Japan."** Coauthored with Minchung Hsu, Pei-Ju Liao, and Yi-Chan Tsai.

   Population aging results in the problem of labor shortage. The labor force of Japanese females presents a solution to the Japanese government. Despite measures to improve the female labor force participation rate, encouraging women to work more or even shift from part-time jobs to full-time jobs was considered by the government. This study employs a standard incomplete markets model (or the Bewley-Huggett-Aiyagari style model) with heterogeneous households to examine the impact of two tax reforms (2004 and 2018) and a counterfactual reform abolishing spousal deduction on women's employment. Findings indicate that reducing spousal deduction after the 2004 tax reform boosts labor participation due to negative wealth effects, while the 2018 income threshold extensions result in a shift from full-time to part-time employment. Conversely, completely abolishing spousal deduction encourages a shift from part-time to full-time employment.

3. **"A Two-Stage Pseudo-Maximum Likelihood Method for Poisson Models with Two Binary Endogenous Explanatory Variables."** Coauthored with Chingmu Chen, Shin-Kun Peng, and Wen-Jen Tsay.

   This paper introduces a novel estimator for the Poisson model with dual binary endogenous explanatory variables, addressing the need to quantify the effects of preferential economic integration agreements. We develop a two-stage pseudo-maximum likelihood estimator and derive analytical gradient matrices as well as Hessian matrices. These exact formulas improve computational speed significantly. Applying our approach to trade data, we find that preferential trade agreements have a positive impact on bilateral trade flows, while bilateral investment treaties have a negative effect. Additionally, the positive interaction term between these two treaties suggests that preferential trade agreements promote horizontal foreign direct investment.

4. **"Robust Inference for Interrupted Time Series Models with Serially Dependent Disturbances."** Coauthored with Wen-Jen Tsay and Hsin-Chieh Wong. *Preparing for submission*.

   This paper considers inference problems related to the interrupted time series (ITS) model, which is used as a quasi-experimental approach for evaluating policy intervention effects over time. Under a single-treatment-group design with two interventions and serially correlated errors, we establish the asymptotic distribution of the ordinary least squares (OLS) estimators, which involves Brownian motion functionals and reveals a super-consistency property for the trending variables when the estimation error is properly rescaled. Interestingly, the asymptotic covariance matrix of the rescaled estimator is proved to be invertible, even though the convergence rates of the OLS estimator for the level and trending variables differ, and the trending variables have different observation lengths. We further develop a studentized sieve bootstrap procedure and establish its asymptotic validity. In simulation experiments, the proposed bootstrap reduces the size distortion of t-tests based on standard normal critical values in every configuration considered.
   doi:10.2139/ssrn.5486489

5. **"A Two-Step Poisson Quasi-Maximum Likelihood Estimator for Structural Poisson Models with Multiple Binary Endogenous Regressors."** Coauthored with Wen-Jen Tsay.

   This paper studies structural Poisson models for count outcomes with multiple endogenous binary regressors. While the existing literature mainly considers cases with one or two endogenous binary variables, we extend the models to allow for k binary endogenous explanatory variables and develop a two-step Poisson quasi-maximum likelihood estimator for this setting. We show that the conditional mean admits a representation as a ratio of multivariate normal probabilities, which makes the structural model estimable in practice. We then derive analytical score and Hessian formulae of the corresponding likelihood function from recursive multivariate normal distribution derivatives and establish a Murphy-Topel-type asymptotic variance formula for valid inference. Monte Carlo results show that the proposed estimator can substantially reduce bias in the main structural coefficients relative to naive Poisson quasi-maximum likelihood when endogeneity is strong.

6. **"Empirical Relevance of Collateral-based and Cash Flow-based Credit Constraints in the U.S. Economy."**

   This study explores the significance of cash flow-based borrowing constraints, testing their empirical relevance across various settings. It investigates whether these constraints can explain key features of U.S. economic data, such as the lead-lag relationship and volatility of residential and nonresidential investments. The study also compares cash flow-based constraints with asset-based counterparts to determine which is more empirically relevant. Finally, it examines the primary driving force behind the Great Recession, focusing on the impacts of deteriorating credit conditions on households and firms. The aim is to provide a clearer understanding of the main factors responsible for the downturn, informing future policy decisions.

CONFERENCE PRESENTATION

National Taipei University Interschool Academic Conference — 2026, Taipei, Taiwan, January 19, 2026.
"A Two-Stage Pseudo-Maximum Likelihood Method for Poisson Models with Two Binary Endogenous Explanatory Variables: An Application to Trade and Investment Agreements."

Taiwan Economic Association Annual Conference — 2025, Taipei, Taiwan, November 29, 2025.
"Asymptotic Distribution of OLS Estimator for Interrupted Time Series Model."

Macroeconometric Modelling Workshop — 2024, Taipei, Taiwan, December 11–12, 2024.
"The Labor Supply of Married Women and Spousal Tax Deductions in Japan."

Macroeconometric Modelling Workshop — 2023, Taipei, Taiwan, November 27–28, 2023.
"Financial Frictions and Uncertainty Shocks."

Taiwan Economic Association Conference — 2016.
"Estimating the Willingness to Pay for the Civic Value When Absentee Voting Is Not Allowed."

Discussant

Discussant, "Overnight Momentum and Intraday Reversal: Evidence from Taiwan Listed Companies," National Taipei University Interschool Academic Conference, 2026.

INVITED SEMINARS

Fu Jen Catholic University, 2026

National Sun Yat-sen University, 2025

Fu Jen Catholic University, 2025

Soochow University, 2024

TEACHING EXPERIENCE

National Dong Hwa University – Instructor

Macrofinance Fall 2026

Macroeconomics (I) Fall 2026

Econometric Analysis (II) Spring 2026

Introduction to Big Data Analytics Fall 2025

Development Economics Fall 2025

Financial Markets Spring 2025–2026

Financial Economics (I) Spring 2025

National Taiwan University – Teaching Assistant

Macroeconomic Theory (I) Fall 2019–2022

Macroeconomic Theory (II) Spring 2020–2021, 2024

Theory of Income and Employment (III) Fall 2022

Economics: Empirical Study and Forecasting (I & II) 2019–2022

Basic Macroeconomics Summer 2022

Basic Quantitative Method Summer 2020–2022

Principle of Economics (I & II) 2019–2020